

Chapter 12

Conventional Sources of Energy

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

Which types of resources of energy are coal, petroleum and natural gas?

1. conventional sources
2. non-renewable sources
3. non-conventional sources
4. Both (a) and (b).

Answer

Both (a) and (b) i.e., conventional sources & non-renewable sources

Question 2

Coal is a solid stratified rock of organic and mineral matter.

1. combustible
2. non-combustible
3. renewable
4. None of the above.

Answer

combustible

Question 3

..... coal has per cent of carbon content and is the most preferred variety of coal.

1. Lignite, 40
2. Peat, 50

3. Anthracite, 90
4. Bituminous, 80

Answer

Anthracite, 90

Question 4

Bituminous coal is also known as

1. coking coal
2. Peat
3. Steam oven
4. Gas stove

Answer

coking coal

Question 5

Anthracite is ideal for domestic use as it is smokeless and has high value.

1. industrial
2. calorific
3. volatile
4. demand

Answer

calorific

Question 6

..... is also known as brown coal and is lower grade coal.

1. Peat
2. Lignite
3. Anthracite
4. Bituminous

Answer

Lignite

Question 7

..... is the oldest coalfield in India.

1. Raniganj
2. Jharia
3. Korba
4. Singrauli

Answer

Raniganj

Question 8

Gondwana coal has variable quantities of

1. Low moisture
2. Small sulphur
3. Small variable quantity of phosphorus
4. All of the above

Answer

All of the above

Question 9

The Neyveli Lignite coal field in is the largest lignite deposit field in South India.

1. Andhra Pradesh
2. Telangana
3. Puducherry
4. Tamil Nadu

Answer

Tamil Nadu

Question 10

The liquid petroleum is called

1. Crude oil
2. Liquid oil
3. Natural gas
4. Natural oil

Answer

Crude oil

Question 11

Light distillates : gasoline :: middle distillates :

1. diesel
2. Lubricating oils
3. naphtha
4. LPG

Answer

diesel

Question 12

Petroleum is used :

1. as a fuel
2. as raw material
3. for power generation
4. All of the above

Answer

All of the above

Question 13

Name the main component of LPG.

1. Methane
2. Butane
3. Propane
4. Ethane

Answer

Butane

Question 14

..... is the first refinery in the private sector in India.

1. The Reliance Petroleum Ltd.
2. The Vadinar Oil Refinery
3. Bina Oil Refinery
4. Panipat Oil Refinery

Answer

The Reliance Petroleum Ltd.

Answer the following questions

Question 1(a)

What is meant by conventional sources of energy? Name any two conventional sources of energy.

Answer

Conventional sources of energy are those which have been used by man for a long time and are still being tapped and used abundantly.

These include coal, petroleum, natural gas and hydel power.

Question 1(b)

Name two main coal bearing areas in India.

Answer

Two main coal bearing areas in India are Raniganj in West Bengal and Jharia in Jharkhand.

Question 1(c)

Give a geographical reason for each of the following:

- (i) Anthracite is used for domestic purposes.
- (ii) Oil refineries are located close to oilfields or near ports.

(iii) The location of coalfields is an important factor in industrial development.

Answer

(i) Anthracite is used for domestic purposes as it is a smokeless fuel with high calorific value.

(ii) Oil refineries are located close to oilfields or near ports to minimise the cost of transport and to avoid the risk of transporting the oil inside the country due to its inflammable nature.

(iii) Location of coal fields is an important factor in industrial development because it is the basic source of power and it is heavy and bulky also, hence difficult to transport.

Question 1(d)

(i) Name the varieties of coal found in India.

(ii) Give a characteristic of each type of coal.

Answer

(i) The varieties of coal found in India are anthracite, bituminous, lignite and peat.

(ii) Characteristic of each type of coal are:

1. **Anthracite coal** — It is shiny and lustrous, almost jet black.
2. **Bituminous coal** — It is hard, black and compact and makes up almost 80% of the world's total coal output.
3. **Lignite coal** — It comprises 40% of carbon and a good deal of moisture and less of combustible matter.
4. **Peat coal** — It has the least carbon content and is inferior to the other three varieties of coal.

Question 2(a)

How is Gondwana coalfield different from tertiary coalfield in terms of location and quality?

Answer

Gondwana Coalfields	Tertiary Coalfields
Gondwana coalfields are largely confined to river valleys like those of Damodar, Mahanadi and Godavari. These coalfields are located in the states of West Bengal, Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh, Maharashtra, Uttar Pradesh, Andhra Pradesh and Telangana.	Tertiary coalfields are found in Arunachal Pradesh, Meghalaya, Kashmir and in the coastal areas of Gujarat and inland basin of Rajasthan.

Gondwana Coalfields	Tertiary Coalfields
This coal is free from moisture and contains sulphur and phosphorus in small variable quantities.	This coal has a higher moisture content, contains more sulphur, with less calorific value.

Question 2(b)

Name the coalfield that is oldest and one that is largest in India.

Answer

The oldest coalfield in India is in Raniganj in West Bengal and the largest coalfield is in Jharia in Jharkhand.

Question 2(c)

Give a geographical reason for each of the following:

(i) Petroleum is called 'liquid gold'.

(ii) Natural gas is a fossil fuel.

(iii) CNG is an eco-friendly fuel.

Answer

(i) Petroleum is called 'liquid gold' because there is not even a tiniest part of crude petroleum which goes to waste or remains unused.

(ii) Natural gas is a fossil fuel because it has been formed by decomposing remains of dead animals and plants buried under the earth.

(iii) CNG is an eco-friendly fuel because it causes less damage to the environment as it is made up of methane and results in less carbon emissions.

Question 2(d)

(i) State two advantages of coal as a source of power.

(ii) Where is the oldest oilfield located in India?

Answer

(i) Two advantages of coal as a source of power are-

1. Energy produced from coal fired plants is cheaper and more affordable than other energy sources.
2. Coal is easy to burn and produces high energy upon combustion.

(ii) The oldest oilfield in India is Digboi oilfield situated in Tinsukhia district of Assam.

Question 3(a)

What is Natural Gas? Name two variants of natural gas.

Answer

Natural gas is a fossil fuel which occurs in association with mineral oil. It is mainly composed of methane with small amounts of propane and ethane.

Two variants of natural gas are-

1. Compressed Natural Gas
2. Liquefied Petroleum Gas

Question 3(b)

Where is natural gas found in India? Name two coastal and two inland oil-refineries in India.

Answer

Over three-fourths of India's natural gas comes from Mumbai High. The rest is produced in Assam, Tamil Nadu, Rajasthan and Tripura.

Two coastal oil-refineries in India are-

1. Haldia refinery – Indian Oil Corporation Limited in Haldia, West Bengal
2. Visakhapatnam Refinery – Hindustan Petroleum Corporation Limited in Visakhapatnam, Andhra Pradesh

Two inland oil-refineries in India are-

1. Bina Refinery – Bharat Oman Refineries Limited in Bina, Madhya Pradesh.
2. Panipat Refinery – Indian Oil Corporation Limited in Panipat, Haryana.

Question 3(c)

(i) Why do the natural gas deposits mostly accompany oil deposits?

(ii) What is added to domestic gas cylinders, which gives a foul smell? Why?

Answer

- (i) Natural gas deposits mostly accompany oil deposits because it has been formed by decomposing remains of dead animals and plants buried under the earth.
- (ii) Ethyl mercaptan, a foul smelling gas is added to domestic gas cylinders, so that any leakage of LPG from the cylinder can be detected instantaneously.

Question 3(d)

Give two advantages and one disadvantage of natural gas.

Answer

Two advantages of natural gas are-

1. It is a cleaner, cheaper fuel than diesel or gasoline.
2. Natural gas is an eco-friendly fuel.

One disadvantage of natural gas is that leaks of natural gas are difficult to detect and are very dangerous.

Question 4(a)

Name one important area that has large coal deposits in:

- (i) Jharkhand
- (ii) West Bengal

Answer

- (i) Jharia has large coal deposits in Jharkhand.
- (ii) Raniganj has large coal deposits in West Bengal.

Question 4(b)

State any one disadvantage of coal found in India.

Answer

One disadvantage of coal found in India is that its calorific value is low.

Question 4(c)

- (i) Name two states that have deposits of the Gondwana coal.

(ii) Name two industries that use large quantity of coal.

Answer

(i) Jharkhand and Odisha have deposits of the Gondwana coal.

(ii) Two industries that use large quantity of coal are-

1. Iron and Steel industries
2. Thermal energy power plants

Question 4(d)

(i) Which state is the largest producer of mineral oil?

(ii) Mention any two advantages of hydel power.

Answer

(i) Assam is the largest producer of mineral oil.

(ii) Two advantages of hydel power are-

1. Hydel power is clean, non-polluting source of energy.
2. It does not produce any Greenhouse gases and prevents global warming.

Question 5(a)

Why does India have to import oil?

Answer

India has to import oil to meet the ever increasing industrial and domestic demands.

Question 5(b)

Name one refinery which belongs to the private sector and one in the public sector.

Answer

Jamnagar refinery in Gujarat belongs to the private sector and Guwahati Refinery in Assam is a public sector refinery.

Question 5(c)

Name the two coastal oil-refineries, one on the west coast of India and the other on the east coast of India. Name one inland oil refinery. List the sources of crude oil for these refineries.

Answer

Below are the oil-refineries along with their location and source of crude oil:

1. Oil refinery — Trombay refinery in Maharashtra
Location — West coast of India
Source of crude oil — Ankleshwar fields
2. Oil refinery — Visakhapatnam Refinery in Andhra Pradesh
Location — East coast of India
Source of crude oil — Tatipaka field
3. Oil refinery — Mathura refinery in Uttar Pradesh
Location — Inland oil refinery
Source of crude oil — Mumbai High

Question 5(d)

Where are the deposits of Petroleum normally located? Give two uses of Petroleum besides being used as a fuel.

Answer

Petroleum is normally found in underground reservoirs in sedimentary rock formations like sandstone, shale and limestone. The entire production of India comes from the Assam-Arakan belt, the Gujarat-Cambay belt and the Mumbai High offshore zone.

Two uses of Petroleum besides being used as a fuel are-

1. Petroleum after refining is used as raw material for the production of various petrochemical products like synthetic rubber, synthetic fibre, polysterene, PVC, phenol, gasoline etc.
2. It is used for power generation.

Question 6(a)

Name four products that are obtained during the refining process of petroleum.

Answer

Four products that are obtained during the refining process of petroleum are-

1. Petrol
2. Diesel
3. Kerosene
4. Tar

Question 6(b)

State any two advantages of using petroleum as a source of power.

Answer

Two advantages of using petroleum as a source of power are-

1. It is a stable energy source.
2. It is a high density fuel.

Question 6(c)

State any three disadvantages of using petroleum.

Answer

Three disadvantages of using petroleum are-

1. Petroleum is a natural fossil fuel and non-renewable.
2. Extracting and burning petroleum generates greenhouse gases that contribute to global warming and environmental pollution.
3. It is expensive because of its limited supply and high demand.

Question 6(d)

State any three disadvantages of hydel power.

Answer

Three disadvantages of hydel power are-

1. The cost of building dams for producing hydroelectricity is quite high.
2. Building a large dam alters the natural water table level.
3. Building dams displaces many people from the area and disturbs their life physically, mentally and psychologically.

Question 7(a)

Where is Mumbai High? What is Sagar Samrat?

Answer

In Maharashtra, about 176 km off Mumbai in the Arabian Sea, there is a huge oil deposit known as Mumbai High.

The syncline of the rock structure in which oil has been struck is higher than the normal height and oil is taken from the depth of over 1,400 m with the help of a specially designed platform known as 'Sagar Samrat'. It is a self propelled jack-up type of drilling platform.

Question 7(b)

Why is coal often used near the source of its mining whereas mineral oil is transported to great distances?

Answer

Coal is a bulky material and its transportation is difficult and costly. Secondly, coal is not a hard but a soft mineral. If not transported in a proper way, it may get crushed and reduced to ash. This results in reduction of the weight of coal. Hence it is used near the source of mining.

Oil is transported to greater distances through pipelines which is a cheaper and easier means of transport.

Question 7(c)

Name the region in which India's main coalfields are located. What is the quality of coal found here? What are its uses?

Answer

The Gondwana coalfields are the largest in India as it accounts for 98% of the total reserves of coal in India. Gondwana coal fields are located in the states of West Bengal, Jharkhand, Odisha, Chhattisgarh, Maharashtra, Madhya Pradesh, Uttar Pradesh and Andhra Pradesh. These coal fields accounts for four-fifth of India's coal reserves.

Gondwana coal is mainly of bituminous quality. It is almost free from moisture and contains sulphur and phosphorus in small variable quantities.

Gondwana coal is used to produce coke, coal gas and steam coal.

Question 7(d)

State three advantages of dams.

Answer

Three advantages of dams are-

1. Dams built to produce hydroelectricity help to save and restore water.
2. Dams provide water for irrigation.
3. Dams prevent flooding.

Question 8(a)

What is meant by hydel power?

Answer

Electricity generated from water is called hydel power.

Question 8(b)

With reference to the Bhakra Nangal Dam, answer the following:

- (i) Name the river on which it has been constructed.
- (ii) Name two states that benefit from this project.

Answer

- (i) Bhakra Nangal dam has been constructed on the river Sutlej.
- (ii) Punjab and Himachal Pradesh benefit from this project.

Question 8(c)

- (i) On which river is Hirakud Dam located?
- (ii) Give any two uses of this dam.

Answer

- (i) Hirakud dam is built across Mahanadi river.
- (ii) Two uses of this dam are-
 - 1. It regulates the flow of the Mahanadi river and produces hydroelectricity through several hydroelectric plants.
 - 2. The dam helps control floods in the mahanadi delta and irrigates 75,000 sq km of land.

Question 8(d)

State three disadvantages of building a big dam.

Answer

Three disadvantages of building a big dam are-

- 1. The cost of building dams for producing hydroelectricity is quite high.

2. Building a large dam alters the natural water table level.
3. It can cause serious geological damage like triggering earthquakes in the area.

Thinking Skills

Question 1

Suppose you are involved in the printing industry. Which conventional source of energy you would be using and for what? How is this resource affecting the lives of people in India?

Answer

In the printing industry, the conventional source of energy typically used is electricity. It powers the printing press, printers, cutting machines, and other equipment necessary for the printing process.

Electricity affects the lives of people in India in the following ways:

1. Employment — It sustains jobs and livelihoods for workers involved in printing-related tasks, contributing to employment opportunities in India.
2. Economic growth — When printing companies operate efficiently, it supports other sectors such as publishing, packaging, advertising, and education. This fosters economic growth and contributes to the overall development of India.
3. Access to printed materials — Electricity-powered printing ensures access to printed materials that are crucial for education, information dissemination, entertainment and cultural preservation.
4. Communication — Reliable electricity supply enables the production of newspapers and magazines, serving as important sources of news and information for people across India.
5. Advertising and marketing — Electricity powers the production of marketing materials such as banners, brochures, and flyers, which is essential for advertising and marketing activities.
6. Environmental impact — The impact of electricity generation on the environment depends on the energy mix. Transitioning to cleaner and renewable sources of electricity can help mitigate environmental consequences, reducing pollution and greenhouse gas emissions associated with the printing industry.

Question 2

Name one fuel which has been developed as an alternative to petrol and diesel for use in vehicles. Name the fossil fuel from which it is derived and is preferred than the conventional source of energy from which it is derived?

Answer

Compressed Natural Gas (CNG) has been developed as an alternative to petrol and diesel for use in vehicles.

It is derived from Natural gas. It is preferred over other conventional sources of energy such as petrol and diesel because of the following reasons:

1. Natural gas is an eco-friendly fuel as it causes less damage to the environment.
2. It is easier to preserve than other fuels. It can be stored and transported through pipelines, small storage units, cylinders or tankers on land and sea.
3. It can be piped into houses for heating and cooking purposes and running a variety of appliances.
4. It is a cleaner, cheaper fuel than diesel or gasoline.
5. It is lighter than air and tends to dissipate when there is a leakage unlike propane, which being heavier than air, collects into explosive pockets.

Question 3

Imagine that you live in an area where a dam is being constructed. What advantages and disadvantages do you foresee once this hydro-project is completed?

Answer

I foresee the following advantages once this hydro-project is completed:

1. Renewable energy generation — The dam can harness flowing water to generate renewable hydroelectric power, reducing reliance on fossil fuels.
2. Clean energy production — Hydroelectric power produces minimal greenhouse gas emissions and air pollutants, contributing to improved air quality and mitigating climate change.
3. Reliable power supply — The dam provides a consistent and reliable power supply, benefiting industries, households, and other sectors dependent on electricity.
4. Water management and irrigation — Dams can regulate water flow, allowing for water storage and controlled release for irrigation and water management purposes.
5. Positive Environmental impact — It does not produce any Greenhouse Gases and prevents us from the harmful effects of Global Warming.

I foresee the following disadvantages once this hydro-project is completed:

1. Displacement of communities — Construction of the dam may lead to the displacement of local communities, disrupting their livelihoods and cultural practices.
2. Negative Environmental impact — Dams can alter river ecosystems, affecting fish migration, water quality, and downstream habitats. It may also lead to the loss of fertile land due to submergence.

3. Sedimentation and reservoir lifespan — Over time, reservoirs may accumulate sediment, reducing their storage capacity and lifespan.
4. Risk of dam failure — In rare cases, dam failures can occur, leading to catastrophic consequences such as flooding downstream areas.
5. Risk of earthquake — Building a large dam can cause serious geological damage like triggering earthquakes in the area.